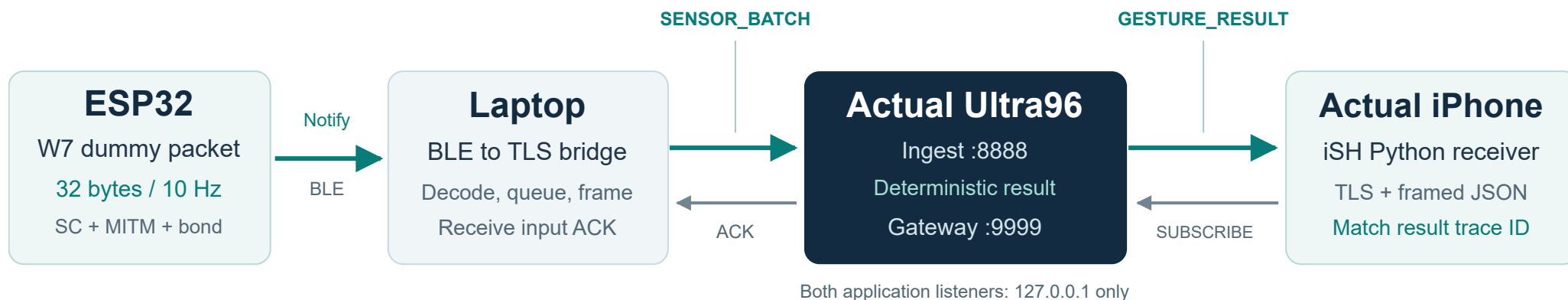


# One ESP. A verified connection path.

Real ESP and Ultra96 hardware deliver 6,100 matching dummy results to the actual iPhone.

Ultra96 external access: TCP 22 (SSH) only

TLS services stay on Ultra96 loopback.



## Laptop owns the ingestion tunnel

Laptop 18888 → Ultra96 8888

Both ends bind 127.0.0.1. The bridge opens TLS through this SSH local forward; the Ultra96 returns **INGEST\_ACK**.

## Phone owns a separate tunnel

Phone 19999 → Ultra96 9999

Both ends bind 127.0.0.1. The Phone opens TLS, sends **SUBSCRIBE**, then receives live **GESTURE\_RESULT**.

Each SSH connection reaches Ultra96 port 22 through the institutional jump host. Host keys are verified; TLS validates the demo CA and `ultra96.week7.internal`.

**Actual Phone: 100/100 and 6,100/6,100.** Latest 100: zero reconnects and checked counters. Long run: one callback discard. Local-forward recovery verified; physical foreground/lifecycle and Unity remain unverified.

# Follow one identity across every hop.

W7 v1 has one sample per notification (**N = 1**). At 10 Hz, the payload is exactly **32 bytes**.

|                           |                      |                      |                                    |                                 |                                        |                                                            |
|---------------------------|----------------------|----------------------|------------------------------------|---------------------------------|----------------------------------------|------------------------------------------------------------|
| 0-1<br><b>W7</b><br>ASCII | 2<br><b>v1</b><br>u8 | 3<br><b>ID</b><br>u8 | 4-7<br><b>boot_id</b><br>uint32 LE | 8-11<br><b>seq</b><br>uint32 LE | 12-15<br><b>uptime_ms</b><br>uint32 LE | 16-31<br><b>8 signed values</b><br>8 × int16 little endian |
|---------------------------|----------------------|----------------------|------------------------------------|---------------------------------|----------------------------------------|------------------------------------------------------------|

Fixed schema: <2sBBIII8h. BLE notification requires ATT MTU ≥ 35; authenticated physical runs measured 517.

ILLUSTRATIVE EXAMPLE / NOT A CAPTURED TRACE

device 1 · boot 7 · seq 42

For channel *i* = 0..7:

```
value[i] = (seq % 2000) - 1000 + 10*i
[-958, -948, -938, -928,
-918, -908, -898, -888]
```

Dummy gesture:

[REST, FIST, OPEN, POINT][seq % 4]  
**42 % 4 = 2** → **OPEN**, confidence 1.0.

Values are dimensionless sentinels. Confidence is a fixed demonstration field, not measured model accuracy.

**Real in this demonstration**  
ESP hardware, authenticated BLE, independent SSH/TLS, actual Ultra96 and iPhone result delivery.

**CORRELATION KEY = DEVICE\_ID:BOOT\_ID:SEQ**  
**1:7:42**  
Configured session: week7-demo  
Boot ID distinguishes device restarts.  
**OPEN**

- 1 Laptop → Ultra96: SENSOR\_BATCH**  
Decode the packet; preserve device, boot and sequence; attach the configured session.
- 2 Ultra96 → Laptop: INGEST\_ACK**  
Accepted input returns the same device, boot and sequence. A recent duplicate emits no second result.
- 3 Ultra96 → viewer: GESTURE\_RESULT**  
Same identifiers, result\_id=1:7:42, **OPEN**. Compare exact ID sets; ACK and result can arrive in either order.

**Dummy by design**  
Sensor values and gestures. Calibrated sensing, AI/FPGA, two-glove synchronization and AR are outside Week 7.

# Show the trace. State the boundary.

Actual iPhone: 6,100 exact results across 609.900 s of source activity; zero receiver reconnects.

## ACTUAL IPHONE / LATEST LONG RUN

# 6,100 / 6,100

Exact Packet / ACK / Phone / board IDs.  
Zero Phone reconnects; six saved exits 0.

Source / ACK spans: 609.900 / 609.781 s.  
One sender callback discard; strict zero-drop check fails. Phone timing unmeasured.

## ACTUAL IPHONE / AFTER RECOVERY

# 100 / 100

Exact IDs :6503 through :6602.  
Zero Phone reconnects; sender 14.078 s.

Zero checked error/drop/gap/callback counters.  
Sender/receiver exits 0; separate capture/wrapper exit files absent.

## PHONE / LOCAL-FORWARD RECOVERY

# 100 / 100

Listener removed, refusal observed,  
TLS restored, then fresh results.

One expected receiver reconnect.  
Receiver restart also verified.  
No SSH-master/VPN or iOS UI loss tested.

## Live show checklist

- 1 **Establish provenance.** VPN connected; verify host trust, actual Ultra96 service and loopback listeners.
- 2 **Open independent SSH forwards.** Laptop 18888 → 8888; Phone 19999 → 9999. Both use Ultra96 SSH port 22. Stop the desktop subscriber.
- 3 **Subscribe before sending.** Capture full Phone output; start the logging BLE sender. Retain retries and every drop counter.
- 4 **Point to one shared trace.** Show device, boot, sequence, accepted ACK and matching result ID in the separate streams.
- 5 **Finish with counted evidence.** Show exact ID-set match, duration and error/drop metrics; close owned clients and forwards.

Use this pack's README and runbooks for commands. If the board path is unavailable, label an offline walkthrough explicitly.

## Pending: physical lifecycle + Unity

**Phone:** source/ACK activity exceeded 600 s; JSONL has no Phone timestamps. Full-run visible foreground confirmation is pending. The long-run display helper covered only the latter 339.476 s. SSH-master/VPN loss and physical app/screen actions remain untested.

**Unity:** target build and teammate integration remain pending. Exact-ID delivery is not full Gate M acceptance.

**Observed physical recovery:** USB-only power loss caused a 21.431 s serial outage; the separate RESET kept serial open. Maximum ACK/result gaps: USB 25.656/25.594 s; RESET 14.000/13.938 s. Both recovered with new boots and matching board-log traces.

**Actual Phone trust:** TLS 1.3 accepted; wrong name / untrusted CA rejected (codes 62 / 20).  
**Earlier desktop soak:** 5,965 exact matches in 600.563 s, with one shutdown callback discard. Historical physical recovery remains desktop evidence.